

Alexander (Xander) Neary

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PROFILE

Software developer and UCLA Computer Science student graduating Dec. 2026, with experience building AR/VR, computer vision, simulation, and real-time interactive systems. Proficient in C++, C#, Python and Unity, with project experience spanning mobile AR, graphics programming, and robotics. Interested in real-time computation with interactive systems.

EXPERTISE

- **Programming Languages:** C++, C#, Python, JavaScript
- **Frameworks & Tools:** Unity, OpenCV, Git, GitHub, Docker, AR Foundation, ARCore, MediaPipe
- **Familiar Technologies:** Arduino, SolidWorks, MuJoCo, AWS, SQL
- **Practices:** Object-Oriented Design, Debugging, Agile/Scrum, Test-Driven Development

PROFESSIONAL EXPERIENCE

Lead Student Instructor – UCLA Engineering 1: Bionics Sep. 2023 – Present

- Led weekly labs for 20+ students covering C++, Arduino, CAD, and hardware integration.
- Mentored and supervised a team of two to three instructors per term, providing guidance on classroom management.
- Diagnosed software-hardware integration issues, reinforcing object-oriented design and system-level debugging skills.

Team Manager – UCLA Esports: Overwatch Main Sep. 2025 – Jun. 2026

- Managed all operations for a competitive collegiate team, coordinating across players, coaches, and league organizers.
- Introduced structured workflows and communication systems to improve preparation efficiency and team coordination.

Software Developer – Summer Undergraduate Research Experience (SURE), USC Jun. 2024 – Aug. 2024

- Designed audiovisual systems in Unity (C#) for a VR simulator supporting chronic pain research studies.
- Engineered procedural wave simulation by tweaking custom shaders, improving visual realism and runtime stability.
- Developed and integrated Unity Editor automation tools, reducing researcher simulation setup time by ~40%.

PROJECTS

ECARUII (Eye-Controlled AR UI Interaction) | Mobile AR App 2026

C#, C++, Unity, AR Foundation, ARCore, MediaPipe, YOLO11, ONNX Runtime, Vulkan – [Repository](#) | [Technical Doc](#)

- Designed and developed an Android AR application combining camera-based eye tracking, nine-point gaze calibration, gaze-controlled UI interaction, YOLO11 object segmentation, and AR world-space placement.
- Built a dual-camera pipeline that maps detected object masks to AR depth queries and persistent anchors while continuously processing front-camera facial landmarks for gaze estimation.
- Developed a native C++ Vulkan-AHardwareBuffer-EGL/GLES interoperability bridge to pass Unity camera textures into MediaPipe's GPU pipeline without CPU readback.

3D Snake | Web Graphics Project 2026

JavaScript, CSS, WebGL, TinyGraphics/Three.js, GitHub – [Repository](#)

- Developed a 3D browser game with frame-independent movement, collision handling, and switchable camera views.
- Built responsive JavaScript/CSS UI systems for game states, scoring, menus, and player interaction.

Robotic Hand-Tracking Control System 2025

Python, OpenCV, MuJoCo, GitHub – [Repository](#)

- Developed a real-time OpenCV hand-tracking pipeline that maps detected hand motion to a simulated robotic arm.
- Integrated vision-derived control inputs with MuJoCo physics and PID controls to produce stable real-time robotic motion.

EDUCATION

University of California, Los Angeles (UCLA) Sep. 2022 – Dec. 2026

Bachelor of Science in Computer Science

Los Angeles, CA

GPA: 3.2

- Relevant Coursework: Robotics, Computer Network Fundamentals, Computer Graphics, Computer Animation
- Organizations: Center for Excellence in Engineering and Diversity (CEED), UCLA Esports

INTERESTS

- Competitive esports, game development, beach volleyball, video editing